

Nick Eisenbergs Comprehensive Presentation

Fall 2020

Gaussian Processes

A Gaussian Processes, $G := \{G_t : t \in T\}$ is a stochastic process such that for any finite set of $t_1, \dots, t_n \in T$, the random vector $(G(t_1), \dots, G(t_n))$ follows a n -dimensional Gaussian distribution. The processes G is uniquely determined by

1. $\mu(t) = \mathbb{E}(G(t))$
2. $C(s, t) = \mathbb{E}(G(t)G(s))$

Theorem (Important)

The space of all symmetric and non-negative definite functions, $f(s, t)$, on $T \times T$ matches the space of all covariance functions of all Gaussian processes on T . Thus, given a non-negative definite function on $T \times T$ then there exists a Gaussian Processes that possesses this covariance function.

Examples of Gaussian Processes

1. **Brownian Motion:** Consider the following mean and covariance functions on $T = [0, \infty)$.

- ▶ $\mu(t) = 0$
- ▶ $C(s, t) = \min(s, t)$

It can be shown that C is non-negative definite and the resulting Gaussian Processes is referred to as Brownian Motion,
 $B = \{B(t) : t \geq 0\}$.

2. **White Noise:** Consider the following mean and covariance functions on $T = \mathcal{B}(\mathbb{R}^N)$, the Borel sets of $\mathbb{R}^N$.

- ▶ $\mu(A) = 0$
- ▶ $C(A, B) = \lambda^N(A \cap B)$, $\lambda^N = N$ -dimensional Lebesgue measure.

Again it can be shown that C is non-negative definite and thus the resulting Gaussian processes on T is referred to as White Noise,
 $\dot{W} = \{\dot{W}(A) : A \in \mathcal{B}(\mathbb{R}^N)\}$.

Brownian Motion in terms of white noise

- ▶ Consider a white noise, $\dot{W} = \{\dot{W}(A) : A \in \mathcal{B}(\mathbb{R})\}$
- ▶ Define $B(t) = \dot{W}([0, t])$
- ▶ Since $\dot{W}$ is a white noise we have
$$\mathbb{E}(B_s B_t) = \mathbb{E}(\dot{W}[0, s] \dot{W}[0, t]) = \lambda([0, s] \cap [0, t]) = \min\{s, t\}$$
- ▶ This shows that B is a Brownian Motion

White Noise as an Ito Integral

We define a set function on $\mathcal{B}(\mathbb{R})$

$$\dot{W}(A) = \int_0^\infty 1_A(t) dB(t, \omega)$$

For any borel set, A , the deterministic indicator function $1_A \in L^2(\mathcal{P})$ and so the Ito Integral exists.

From the theorem in the previous slide we see that

- ▶ $\mathbb{E}\dot{W}(A) = 0$
- ▶ $\mathbb{E}(\dot{W}(A)\dot{W}(B)) = \int_0^\infty 1_A(t)1_B(t)dt = \lambda(A \cap B)$

Thus, $\dot{W}$ is a white noise on $\mathcal{B}(\mathbb{R})$

Ito Integral

We will briefly explain the meaning of $\int_0^\infty u(t, \omega) dB(t, \omega)$, the Ito Integral of u , where B is a Brownian Motion.

Definition (Progressively Measurable)

A process, $u(t, \omega)$, is progressively measurable if u restricted to $\Omega \times [0, t]$ is $\mathcal{F}_t \times \mathcal{B}([0, t])$ measurable where $\mathcal{F}_t = \sigma\{B(s) : 0 \leq s \leq t\}$.

Definition (Progressively Measurable Hilbert space, $L^2(\mathcal{P})$)

We define $L^2(\mathcal{P}) = L^2(\Omega \times [0, \infty), \mathcal{P}, \mathbb{P} \times \lambda)$ where $\mathcal{P}$ is the sigma-algebra generated by all the sets $A \subset \Omega \times [0, \infty)$ such that $1_A(t, \omega)$ is progressively measurable. The norm of this Hilbert Space is
$$\|u\|_{\mathcal{P}}^2 = \mathbb{E} \left(\int_0^\infty u(t, \omega)^2 dt \right)$$

Definition (Simple Processes and their Ito Integral)

A process of the form $u(t, \omega) = \sum_{k=1}^n \phi_k(\omega) 1_{(t_k, t_{k+1}]}(t)$ where ϕ_k is $\mathcal{F}_k$ measurable is called a simple process. Its Ito integral is defined to be
$$\int_0^\infty u(t, \omega) dB(t, \omega) = \sum_{k=1}^n \phi_k(\omega) (B(t_{k+1}, \omega) - B(t_k, \omega))$$

Ito Integral

Theorem

The simple processes are dense in $L^2(\mathcal{P})$. Thus for any processes, $u(t, \omega)$, there exists a sequence of simple processes, $u_n(t, \omega)$ such that $\mathbb{E} \left(\int_0^\infty (u(t, \omega) - u_n(t, \omega))^2 dt \right) \rightarrow 0$.

Definition (Ito Integral of a $L^2(\mathcal{P})$ processes)

For any processes $u(t, \omega) \in L^2(\mathcal{P})$ we define its Ito Integral to be

$$\int_0^\infty u(t, \omega) dB(t, \omega) := \lim_{n \rightarrow \infty} \int_0^\infty u_n(t, \omega) dB(t, \omega)$$

where the limit is with respect to the $L^2(\mathcal{P})$ norm and the u_n 's are from the Theorem.

Theorem

1. $\mathbb{E} \int_0^\infty u(t, \omega) dB(t, \omega) = 0$
2. $\mathbb{E} \left(\int_0^\infty u(t, \omega) dB(t, \omega) \cdot \int_0^\infty v(t, \omega) dB(t, \omega) \right) = \mathbb{E} \left(\int_0^\infty u(t, \omega) v(t, \omega) dt \right)$

Walsh Integral

$$\int_0^t \int_{\mathbb{R}^d} f(s, y) W(ds, dy)$$

- ▶ Generalization of the Ito Integral
- ▶ Defined for two parameter worthy martingale measure integrators.
 - ▶ The white noise processes $W_t(A) = \dot{W}([0, t] \times A)$ is "worthy" and is used in the 1- d SHE and SWE.
 - ▶ A "colored" Gaussian noise will be used in higher dimensions
- ▶ Used to define the solution to the SHE and SWE for two parameter noise forcing terms.

Walsh Integral Construction

- ▶ Simple functions: Let $\mathcal{L}$ be class of linear combinations of processes of the form $f(t, x, \omega) = X(\omega)1_{(a,b]}(t)1_A(x)$ where $X \in \mathcal{F}_a$ and let $\mathcal{P} = \sigma(\mathcal{L})$
- ▶ The integral for a simple processes is

$$(f \cdot M)_t(B) = X(\omega) [M_{b \wedge t}(A \cap B) - M_{a \wedge t}(A \cap B)]$$

Theorem

For $\mathcal{P}$ measurable processes, $f(t, x, \omega)$ that satisfy

$$\int_0^T \iint_{\mathbb{R}^{2n}} |f(x, t)f(y, t)|K(dx, dy, dt) < \infty$$

there exists a sequence of simple processes whose integrals are Cauchy in $L^2(\mathcal{P})$. The limit is the walsh integral of f .

Colored Noise

- ▶ For non-negative and non-negative definite measure Γ define on $\mathcal{S}(\mathbb{R}^d)$

$$\begin{aligned} C(\varphi, \psi) &= \int_0^\infty ds \int_{\mathbb{R}^d} \Gamma(dx) (\varphi(s, \cdot) * \tilde{\psi}(s, \cdot))(x) \\ &= \int_0^\infty ds \int_{\mathbb{R}^d} dx \int_{\mathbb{R}^d} dy \varphi(s, x) f(x - y) \psi(s, y) \end{aligned}$$

- ▶ C is non-negative definite and the resulting 0-mean Gaussian processes is referred to as a colored Gaussian noise, $F = \{F(\varphi) : \varphi \in \mathcal{S}(\mathbb{R}^d)\}$.
- ▶ Bochner's Theorem implies the existence of non-negative measure μ (spectral density) such that

$$\int \Gamma(dx) \varphi(x) = \int \mu(dx) \mathcal{F}\varphi(x)$$

Walsh integral with Colored Noise Integrator

- ▶ Take $\varphi_n \in \mathcal{S}(\mathbb{R}^d)$ such that $\varphi_n \rightarrow 1_A$
- ▶ Can be shown $\mathbb{E}[(F(\varphi_n) - F(\varphi_m))^2] \rightarrow 0$. Call the limit $F(A)$.
- ▶ $M_t(A) := F([0, t] \times A)$ is worthy and can be used as an integrator.
- ▶ Through the construction we can show that

$$F(\varphi) = \int_0^\infty \int_{\mathbb{R}^d} \varphi(s, x) M(dx, ds)$$

- ▶ The Walsh integral for integrator M can be defined for processes satisfying

$$\int_0^\infty ds \int_{\mathbb{R}^d} dx \int_{\mathbb{R}^d} dy \varphi(s, x) f(x - y) \varphi(s, y) < \infty$$

Stochastic Heat Equation

$$\begin{cases} \frac{\partial u}{\partial t}(t, x) - \frac{1}{2} \Delta_x u(t, x) = \sigma(u(t, x)) \dot{M}(t, x) \\ u(0, \cdot) = \mu(\cdot) \end{cases}$$

- The rigorous meaning of the SHE is

$$u(t, x) = \int_{\mathbb{R}^d} G(t, x-y) \mu(dy) + \int_0^t \int_{\mathbb{R}^d} G(t-s, x-y) \sigma(u(s, y)) M(ds, dy)$$

- Informally the $\dot{M}$ is acting like the density (does not exist) of M in the sense that

$$\int_0^t \int_{\mathbb{R}^d} \varphi(s, y) M(ds, dy) = \int_0^t \int_{\mathbb{R}^d} \varphi(s, y) \dot{M}(s, y) ds dy$$

Definition of Invariant Measures

Consider an arbitrary measure space $(E, \mathcal{E})$ and denote $\mathcal{M}_1(E)$ to be the space of all probability measure on $(E, \mathcal{E})$.

Definition (Markovian Transition Function)

We say that $P_t(x, \Gamma)$, $t > 0$, $x \in E$, $\Gamma \in \mathcal{E}$, is a Markovian Transition Function is:

- (i) $P_t(x, \cdot)$ is a probability measure on $(E, \mathcal{E})$ for $t \geq 0$, $x \in E$
- (ii) $P_t(\cdot, \Gamma)$ is a $\mathcal{E}$ -measurable function for $t \geq 0$ and $\Gamma \in \mathcal{E}$
- (iii) $P_{t+s}(x, \Gamma) = \int_E P_t(x, dy) P_s(y, \Gamma)$
- (iv) $P_0(x, \Gamma) = 1_\Gamma(x)$ where $1_\Gamma(x)$ is the indicator function.

Invariant Measures viewed as an average

Definition (Invariant Measure)

A measure $\eta(\cdot) \in \mathcal{M}_1(E)$ is invariant for transition functions $\{P_t(x, \Gamma)\}_{t \geq 0}$ if for any $\Gamma \in \mathcal{E}$ we have that

$$\eta(\Gamma) = \int_E P_t(x, \Gamma) \eta(dx) \quad \text{for any } t \geq 0$$

Recall from Calculus that the average of a function $f : [a, b] \rightarrow \mathbb{R}$ is defined as

$$\text{AVG}(f) = \frac{1}{b-a} \int_a^b f(x) dx$$

Thus, $\eta(\Gamma)$ is the average of the function $x \mapsto P_t(x, \Gamma)$ over $(E, \mathcal{E}, \eta)$.

Main Result

The laws of the solution the the SHE defines Markovian transition functions

$$P_t(\varphi, \Gamma) = \mathcal{L}(u^\varphi(t, \cdot))(\Gamma) = P[\omega \in \Omega : u^\varphi(t, \cdot) \in \Gamma] \quad \Gamma \in \mathcal{B}(L_\rho^2)$$

where $u^\varphi(t, x)$ is the solution to the SHE starting at $\varphi \in L_\rho^2$. We have the following main results

Theorem

Under certain conditions of the spectral density $\hat{f}$ and on the admissible weight $\rho(x)$, there exists a measure η that is invariant for the solution to the stochastic heat equation for any L_ρ^2 initial condition. In other words, for $P_t(\varphi, \Gamma)$ we have that, $\eta(\Gamma) = \int_{L_\rho^2(\mathbb{R}^d)} P_t(x, \Gamma) \eta(dx)$ for $t \geq T > 0$ for any $T > 0$.

Theorem

Under the same assumptions as the theorem above, there exists a measure η that is invariant for the solution to the stochastic heat equation with the $\delta_0(x)$ initial condition.

Importance of the Result/Comparison to Previous Results

- ▶ Translation of the paper [12] from Daprato theory to Walsh Theory.
- ▶ Existence of Invariant Measures for a the SHE with a broader range of initial conditions, in particular measure valued initial conditions such as δ_0 .
 - ▶ [12] has constant 1 in initial condition, Lipschitz diffusion coefficient and 0 drift term
 - ▶ [9] allowed L^2_ρ initial conditions with bounded and Lipschitz coefficients.
 - ▶ [1] Allows unbounded coefficients. The diffusion coefficient is Lipschitz and the drift has a polynomial type growth. The weight is given to be $\rho(x) = (1 + x^2)^{-e/2}$.

Current Project: Fractional SPDE

$$\begin{cases} (\partial^b + \frac{\nu}{2}(-\Delta)^{a/2}) u(t, x) = I_t^r \left[\sqrt{\theta} u(t, x) \dot{W}(x) \right] & x \in \mathbb{R}^d, t > 0 \\ u(0, \cdot) = 1 & \beta \in (0, 1] \\ u(0, \cdot) = 1, \quad u_t(0, \cdot) = 0 & \beta \in (1, 2) \end{cases} \quad (0.1)$$

- ▶ $a \in (0, 2]$, $b \in (0, 2)$, $\nu > 0$, $r > 0$ and $\theta > 0$
- ▶ $(-\Delta)^{a/2}$ refers to the fractional Laplacian [8]
- ▶ ∂^b refers to the Caputo derivative [11]
- ▶ I_t^r is the Riemann-Liouville fractional integral operator.
- ▶ The time independent noise $W := \{W(\varphi) : \varphi \in \mathcal{D}(\mathbb{R}^d)\}$ arises from the covariance function

$$\mathbb{E}(W(\varphi)W(\psi)) = \langle \varphi, \psi \rangle_{\mathcal{H}} = \int_{\mathbb{R}^d} \mathcal{F}\varphi(\xi) \overline{\mathcal{F}\psi(\xi)} \mu(d\xi)$$

Fractional Laplacian:

- **Fractional Laplacian:** One way to define the fractional Laplacian is through the Fourier Transform. It can be shown that

$$\mathcal{F}(-\Delta f)(\xi) = K|\xi|^2 \mathcal{F}f(\xi)$$

where K depends on the definition of the Fourier transform. So for $\psi \in \mathcal{S}(\mathbb{R}^d)$ we define

$$\mathcal{F}(\psi(-\Delta)f)(\xi) = \psi(K|\xi|^2) \mathcal{F}f(\xi)$$

and in particular we can show that if we want to solve

$$(-\Delta^{\alpha/2})u = f \implies u = (-\Delta^{-\alpha/2})f$$

then u is given by $f * R_\alpha$ where R_α is the Reisz Kernel.

$$R_\alpha(x) = \frac{\Gamma((1/2)(d - \alpha))}{\Gamma((1/2)\alpha) 2^\alpha \pi^{d/2} |x|^{d-\alpha}}$$

Fundamental solution to the deterministic FPDE

$$\begin{cases} (\partial^b + \frac{\nu}{2}(-\Delta)^{a/2}) u(t, x) = I_t^r [f(t, x)] & x \in \mathbb{R}^d, t > 0 \\ u(0, \cdot) = \mu & \beta \in (0, 1] \\ u(0, \cdot) = \mu_0, \quad u_t(0, \cdot) = \mu_1 & \beta \in (1, 2) \end{cases} \quad (0.2)$$

$$u(t, x) = \begin{cases} (Z(t, \cdot) * \mu)(x) + (Y \star f)(t, x) & \beta \in (0, 1] \\ (\frac{\partial Z}{\partial t}(t, \cdot) * \mu_0)(x) + (Z(t, \cdot) * \mu_1)(x) + (Y \star f)(t, x) & \beta \in (1, 2) \end{cases}$$

- $Z(t, x) = \pi^{-d/2} t^{[\beta]-1} |x|^{-d} H_{2,3}^{2,1} \left(\frac{|x|^\alpha}{2^{\alpha-1} \nu t^\beta} \middle| \begin{matrix} (1, 1) & ([\beta], \beta) \\ (d/2, \alpha/2) & (1, 1) & (1, \alpha/2) \end{matrix} \right)$
- $Y(t, x) = {}_t D_+^{[\beta]-\beta-\gamma} Z(t, x)$

Reason for constant 1 initial conditions

It can be shown that for the above initial conditions that

- ▶ $(Z(t, \cdot) * 1)(x) = 1$ when $\beta \in (0, 1]$
- ▶ $(\frac{\partial Z}{\partial t}(t, \cdot) * 1)(x) = 1$ when $\beta \in (1, 2)$

In addition, it can be shown that

$$\mathcal{F}Y(ct, \cdot)(\xi) = c^{b+r-1} \mathcal{F}Y(t, \cdot)(c^{b/a} \xi).$$

where as for the Wave equation we have

$$\mathcal{F}G(ct, \cdot)(\xi) = c \mathcal{F}G(t, \cdot)(c\xi)$$

and because of these similar scaling properties, we can almost treat Y and G as the same.

FSPDE Integral Equation

- ▶ The time independence leads us to rigourously defining the FSPDE through the following Skorohod integral equation

$$u(t, x) = 1 + \sqrt{\theta} \int_0^t \left(\int_{\mathbb{R}^d} Y(t-s, x-y) u(s, y) W(dy) \right) ds$$

- ▶ Through Piccard iteration we show that

$$u(t, x) = 1 + \sum_{n \geq 1} \theta^{n/2} I_n(f_n(\cdot, x, t))$$

where

$$f_n(x_1, \dots, x_n, x, t) = \int_{[0, t]^n} G(t - t_n, x - x_n) \cdots G(t_2 - t_1, x_2 - x_1) dt$$

Moment Asymptotics for SWE

For the SWE with time independent noise the following two have been proven in [2]

- ▶ Existence and uniqueness of the solution.
 - ▶ Proven by showing $\mathbb{E}(|u(t, x)|^2) = \sum_{n \geq 0} \theta^n n! \left\| \tilde{f}_n(\cdot, x, t) \right\|_{\mathcal{H}^{\otimes n}}^2 < \infty$
- ▶ Let $t_p = (p - 1)^{\frac{1}{4-\alpha}} t$. The following moment asymptotic has been proven for the SWE for dimensions $0 < \alpha < d \leq 3$

$$\lim_{t_p \rightarrow \infty} t_p^{-\frac{4-\alpha}{3-\alpha}} \log \|u(t, x)\|_p = \theta^{\frac{1}{3-\alpha}} \left(\frac{1}{2}\right)^{\frac{\alpha}{2(3-\alpha)}} \frac{3-\alpha}{2} \left(\frac{2\mathcal{M}^{1/2}}{4-\alpha}\right)^{\frac{4-\alpha}{3-\alpha}}$$

FSPDE

- ▶ Existence and uniqueness of the solution
- ▶ Similar asymptotics

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